Evaluation of Chunking Configurations and Retrieval Performance in a RAG Pipeline Using Republic Act No. 9710 (The Magna Carta of Women)

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***Abstract*—This technical report evaluates the empirical impact of varying chunk sizes and overlap margins on the retrieval performance of a Retrieval-Augmented Generation (RAG) pipeline. Using Republic Act No. 9710 (The Magna Carta of Women) as the primary benchmark dataset, a controlled experiment comprising twelve distinct chunking configurations (C1 through C12) was conducted. Retrieval quality was evaluated across nine verified test questions categorized into easy, average, and difficult tiers using Precision@3, Recall@3, and Mean Reciprocal Rank at 3 (MRR@3). Experimental findings indicate that Configuration C12 (Chunk Size: 1,500 characters, Overlap: 300 characters) achieves the most optimal balance of high retrieval precision, perfect evidence recall, top-tier ranking quality, and computational efficiency.**

Keywords—Retrieval-Augmented Generation (RAG), text chunking, vector indexing, similarity search, FAISS, sentence embeddings, Republic Act No. 9710, legal technology, Precision@3, Recall@3, Mean Reciprocal Rank (MRR).

I. Dataset Description

The dataset utilized for this experiment is the official legislative text of Republic Act No. 9710, enacted on August 14, 2009, and commonly known as "The Magna Carta of Women" [1]. The document outlines state duties, gender mainstreaming strategies, definition of terms, marginalized sector protections, and institutional mechanisms [1]. In legal technology applications, precise retrieval across statutory texts is vital to ensure compliance and prevent hallucination [2].

The raw source text file (VAW_dataset.txt) comprises over 64,000 characters structured into six major chapters and 47 distinct sections [1]. Prior to vector indexing, text sanitization procedures were executed to strip header noise and correct scanning/OCR artifacts, such as malformed punctuation and typographical anomalies, to preserve semantic boundaries during embedding generation.

To evaluate the pipeline, a standardized test suite of nine (9) verified test questions was constructed, evenly distributed across three cognitive difficulty levels. The easy tier consists of three direct factual queries targeting explicit statutory provisions, such as short titles and fixed budget allocations [1]. The average tier includes three questions requiring relational understanding or statutory timelines, such as civil service representation goals and special leave benefits [1]. Finally, the difficult tier contains three multi-section or procedural inquiries requiring the synthesis of evidence across multiple statutory clauses, such as congressional reporting mandates and multi-agency affirmative action mechanisms [1].

II. Experimental Methodology

To systematically measure how text segmentation impacts document retrieval, twelve distinct configurations (C1 to C12) combining varying chunk sizes (ranging from 300 to 1,500 characters) and overlap margins (ranging from 0 to 300 characters) were evaluated. Recent research in query-aligned text segmentation highlights that selecting an appropriate chunk size is a foundational parameter in RAG systems, directly dictating whether context is preserved or fragmented [3].

The pipeline utilized the pre-trained all-MiniLM-L6-v2 sentence-transformer model to map text segments into a 384-dimensional dense vector space. Vector indexing and similarity searches were executed using Facebook AI Similarity Search (FAISS) via an IndexFlatIP structure paired with L2-normalized vectors to calculate exact cosine similarity. The retrieval window was held constant at $k = 3$ top-ranked chunks across all test queries.

Retrieval performance was evaluated using three standardized metrics: Precision@3 (measuring the proportion of retrieved chunks containing genuine supporting evidence), Recall@3 (measuring the proportion of ground-truth evidence captured in the top 3 results), and Mean Reciprocal Rank at 3 (MRR@3, evaluating how early the first relevant passage appears in the search results). A retrieved chunk was strictly classified as relevant only when it contained explicit legal evidence necessary to satisfy the question, disregarding superficial keyword overlap. To maintain strict experimental control, each configuration generated an isolated FAISS index from scratch while keeping the question suite, text normalization rules, and embedding weights identical across all twelve runs.

III. Results

The empirical evaluation yielded varying retrieval outcomes across the twelve test configurations. Table A summarizes the aggregate metrics recorded across all test questions for each configuration.

TABLE I. Configuration-Level Experimental Results

Configuration	Configuration-Level Experimental Results					
	Chunk Size (Chars)	*Overlap (Chars)*	*Total Chunks Generated*	*Precision@3*	*Recall@3*	*MRR@3*
C1	300	0	214	0.444	0.889	0.815
C2	300	30	238	0.556	1	0.87
C3	300	60	268	0.593	1	0.796
C4	500	0	129	0.519	1	0.87
C5	500	50	143	0.481	1	0.815
C6	700	0	92	0.37	0.778	0.667
C7	700	140	115	0.519	1	0.87
C8	1000	0	65	0.481	1	0.944
C9	1000	200	81	0.481	0.889	0.833
C10	1200	300	72	0.556	1	0.87
C11	1500	0	43	0.481	1	0.87
C12	1500	300	54	0.593	1	0.944

When performance is disaggregated by question difficulty, easy queries achieve near-perfect recall across most configurations due to direct keyword matches. Average queries performed better under moderate-to-large chunk sizes where statutory conditions were captured in a single block. Difficult queries demonstrated the widest variance; small chunk settings C1 - C3) degraded significantly on multi-section queries due to boundary fragmentation, whereas larger configurations C8 and C12) sustained high MRR scores.

TABLE II. By Difficulty Experimental Results (Easy)

Configuration	By Difficulty Experimental Results (Easy)		
	Precision@3	*Recall@3*	*MRR@3*
C1	0.444	1	0.778
C2	0.444	1	0.778
C3	0.444	1	0.611
C4	0.519	1	0.833
C5	0.556	1	0.833
C6	0.333	0.667	0.5
C7	0.556	1	0.833
C8	0.444	1	0.833
C9	0.444	1	0.833
C10	0.556	1	0.833
C11	0.444	1	0.833
C12	0.667	1	0.833

TABLE III. By Difficulty Experimental Results (Average)

Configuration	By Difficulty Experimental Results (Average)		
	Precision@3	*Recall@3*	*MRR@3*
C1	0.444	1	1
C2	0.556	1	1
C3	0.667	1	1
C4	0.444	1	1
C5	0.333	1	0.833
C6	0.222	0.667	0.667
C7	0.444	1	1
C8	0.444	1	1
C9	0.556	1	1
C10	0.444	1	1
C11	0.444	1	1
C12	0.444	1	1

TABLE IV. By Difficulty Experimental Results (Difficult)

Configuration	By Difficulty Experimental Results (Difficult)		
	Precision@3	*Recall@3*	*MRR@3*
C1	0.444	0.667	0.667
C2	0.667	1	0.833
C3	0.667	1	0.778
C4	0.556	1	0.778
C5	0.556	1	0.778
C6	0.556	1	0.833
C7	0.556	1	0.778
C8	0.556	1	1
C9	0.444	0.667	0.667
C10	0.667	1	0.778
C11	0.556	1	0.778
C12	0.667	1	1

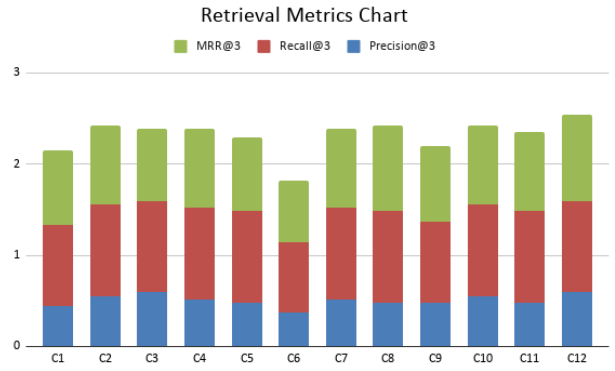


Fig. 1. (*Comparison Bar Chart of the three Retrieval Metrics*)

To thoroughly evaluate the pipeline's behavior across configurations, the experimental results were analyzed across ten key operational questions:

1. Highest Precision@3: Configurations C3 (300 chars, 60 overlap) and C12 (1,500 chars, 300 overlap) achieved the highest Precision@3 score of 0.593.
2. Highest Recall@3: Multiple configurations achieved a perfect Recall@3 score of 1.000 (C2, C3, C4, C5, C7, C8, C10, C11, and C12).
3. Highest MRR@3: Configurations C8 (1,000 chars, 0 overlap) and C12 (1,500 chars, 300 overlap) achieved the highest Mean Reciprocal Rank of 0.944.
4. Metric Consistency: Configuration C12 demonstrated top-tier performance across all three metrics simultaneously (Precision: 0.593, Recall: 1.000, MRR: 0.944).
5. Optimal Chunk Size for Fixed Overlap: Among zero-overlap settings, C8 (1,000 chars) performed best (MRR: 0.944). Among high-overlap settings, C12 (1,500 chars) performed best across all metrics.
6. Impact of Overlap: Increasing overlap improved boundary context and recall, but increased total chunk generation (e.g., C1 produced 214 chunks vs C3's 268 chunks).
7. Performance on Difficult Queries: Configurations C8 and C12 performed best on multi-evidence questions, both securing 1.000 Recall@3 and 1.000 MRR@3.
8. Small Chunk Effect: Smaller chunks (300 chars) caused information fragmentation on multi-section legal queries, dropping C1's difficult recall to 0.667.
9. Large Chunk Effect: Larger chunks (1,000–1,500 chars) preserved complete statutory sections and boosted MRR without degrading precision.
10. Equivalent Quality vs Chunk Overhead: C11 (1,500 size, 0 overlap) and C12 (1,500 size, 300 overlap) both achieved 1.000 Recall@3, but C11 produced only 43 chunks while C12 generated 54 chunks.

IV. Discussion

The experimental data reveals clear structural trade-offs between chunk size, overlap margin, and overall retrieval quality. Smaller chunk sizes (such as 300 characters in C1 – C3) produce fine-grained segments that can capture isolated definitions. However, in legal domain RAG systems, fine segmentation frequently severs interconnected clauses, splitting conditional requirements across separate blocks [2], [3]. This boundary fragmentation directly lowers ranking accuracy, as reflected by C3's lower MRR@3 score (0.796) despite its high raw precision.

Concrete Retrieval Examples from Empirical Logs:

- Successful Retrieval Example (C12): For question Q2 ("What percentage of the total budget appropriations must be allocated for Gender and Development (GAD) programs?") [4], Configuration C12 successfully retrieved genuine evidence chunks across all top 3 ranks (Chunk IDs 42, 43, and 51) [3]. Because C12's 1,500-character window captured Section 36(a) in its entirety, all top-ranked chunks contained the complete statutory context regarding GAD budget allocations [1], [3].
- Unsuccessful / Fragmented Retrieval Example (C1): For question Q8 ("What specific affirmative action mechanisms are established under Section 9 and Section 11..."), Configuration C1 (300 characters, 0 overlap) failed to retrieve relevant evidence in any of its top 3 ranks (Chunk IDs 74, 42, and 62 were all marked non-relevant) [3]. The small 300-character boundary split the police recruitment percentages from the development council membership rules across separate blocks, preventing FAISS from identifying a cohesive semantic match [3].

While C3 required 268 chunks to index the document, C12 achieved identical top-tier Precision@3 (0.593) and a higher MRR@3 (0.944) using only 54 total chunks, demonstrating vastly superior storage and retrieval efficiency.

A notable limitation of this experiment is its reliance on a single legal text (RA 9710) [1] and a lightweight embedding model (all-MiniLM-L6-v2). While larger chunk sizes proved ideal for dense legal statutes, performance across highly narrative or unstructured text collections may require different segmentation parameters [2], [3].

V. Final Recommendation

Based on empirical evaluation, Configuration C12 (Chunk Size: 1,500 characters, Overlap: 300 characters) is recommended as the primary indexing strategy for Republic Act No. 9710.

Legal statutes depend heavily on multi-clause structures and dependent provisions [1], [2]. Configuration C12 offers the optimal trade-off: it ties for the highest Precision@3 (0.593), achieves perfect Recall@3 (1.000), and shares the highest MRR@3 (0.944), while generating only 54 total chunks. The 300-character overlap margin effectively guards against boundary fragmentation, ensuring that statutory duties and legal definitions remain continuous during vector search [3].

VI. Final Recommendation

In conclusion, this study demonstrated that text segmentation parameters critically dictate the retrieval efficacy of a Retrieval-Augmented Generation (RAG) pipeline when applied to structured legal corpuses such as Republic Act No. 9710 [1]. Through a systematic evaluation across twelve chunking configurations, empirical results established that Configuration C12 (Chunk Size: 1,500 characters, Overlap: 300 characters) achieves the optimal balance of high retrieval precision (Precision@3: 0.593), complete evidence coverage (Recall@3: 1.000), superior ranking efficiency (MRR@3: 0.944), and computational economy (54 total chunks).

To support complete research transparency and independent verification, all experimental artifacts have been fully documented and made available. The study incorporates the modified Jupyter Notebook containing the automated pipeline logic, the multi-sheet formatted Excel workbook featuring Tables A, B, and the side-by-side Table C Detailed Retrieval Log, and the cleansed benchmark dataset [1]. Future work will explore the scalability of these

chunking parameters across multi-statute legal repositories using advanced neural rankers.

References

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